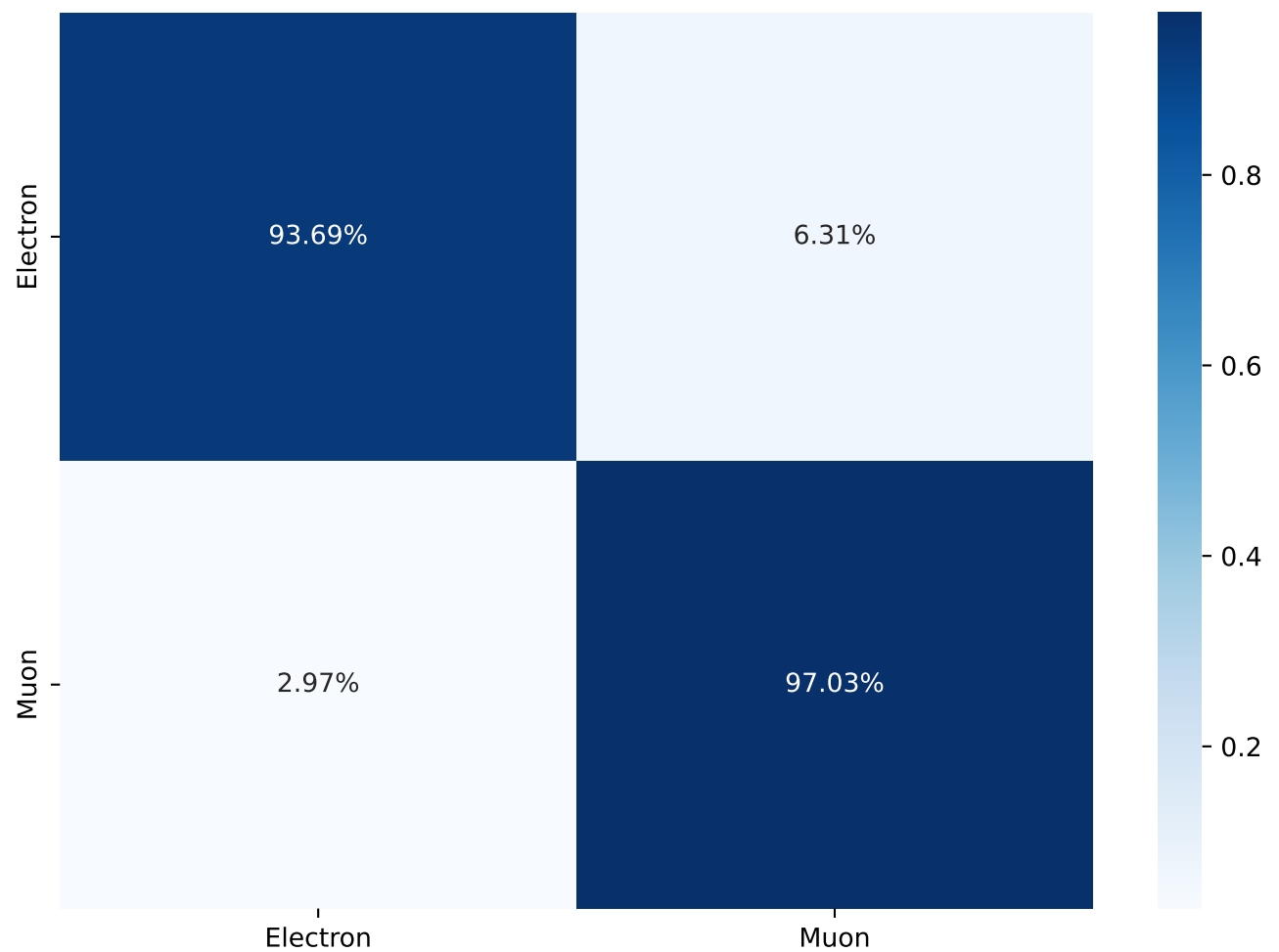
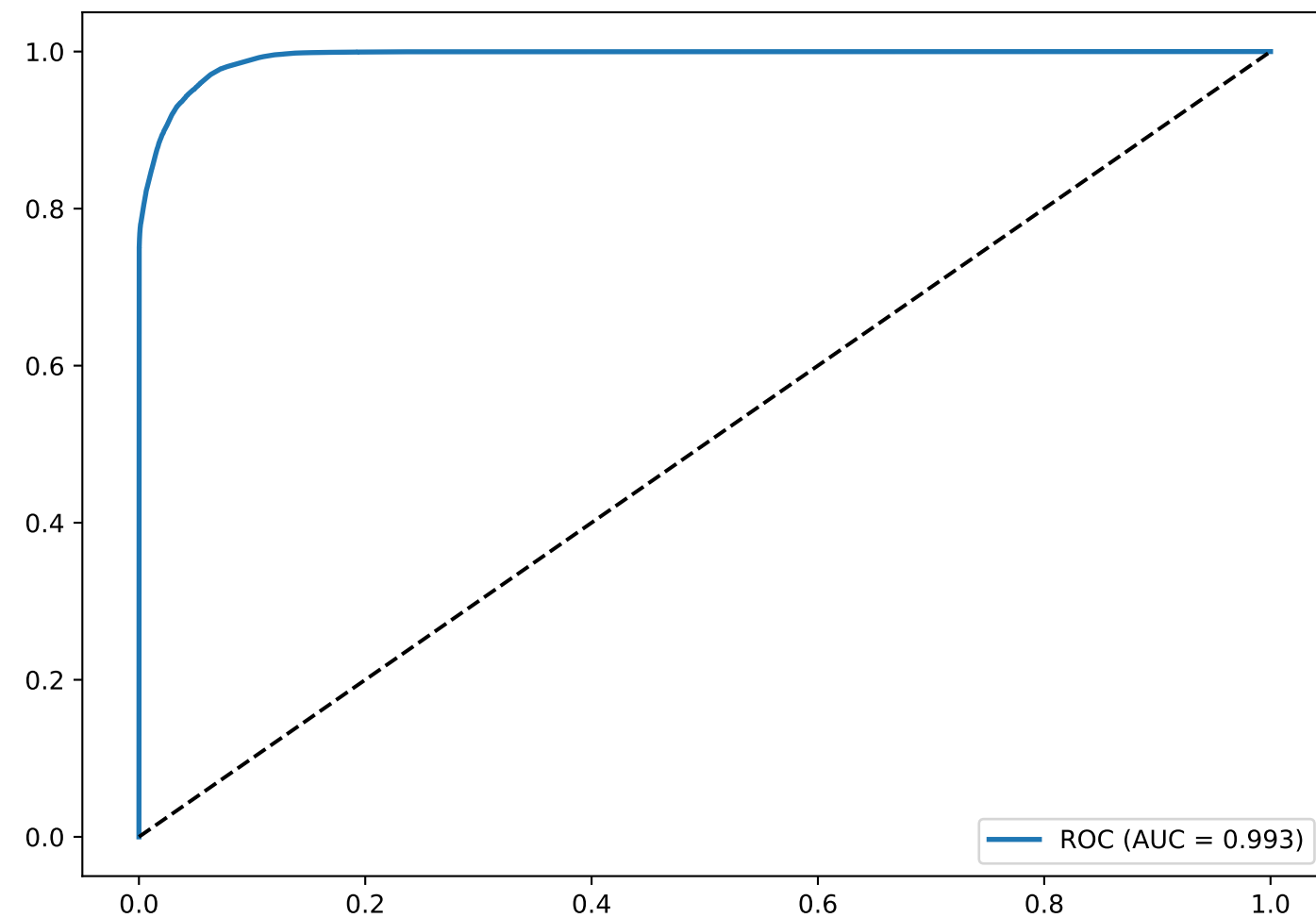


EtaRange = FullRange (TEST DATA) - Accuracy: 0.963

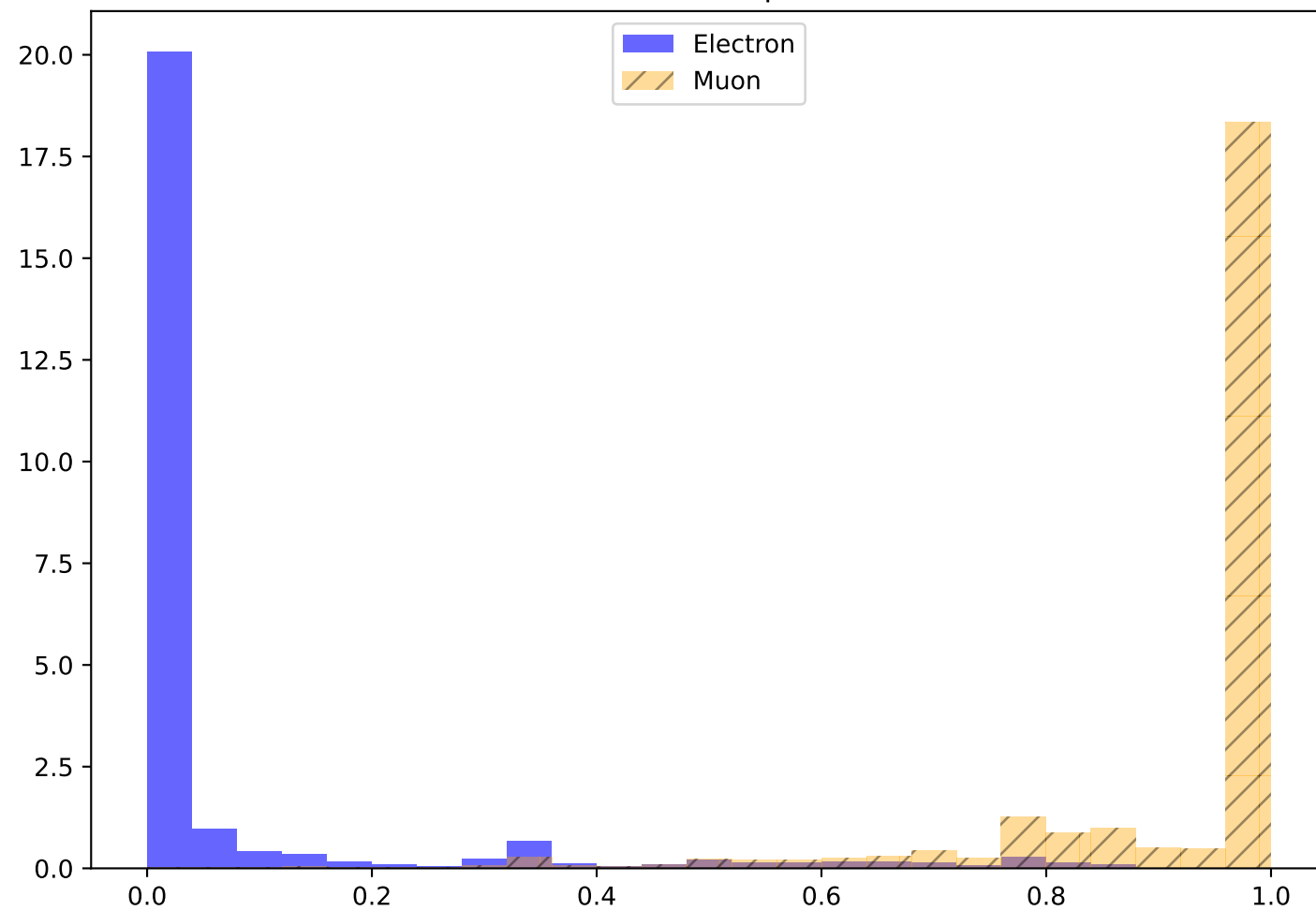
Confusion Matrix



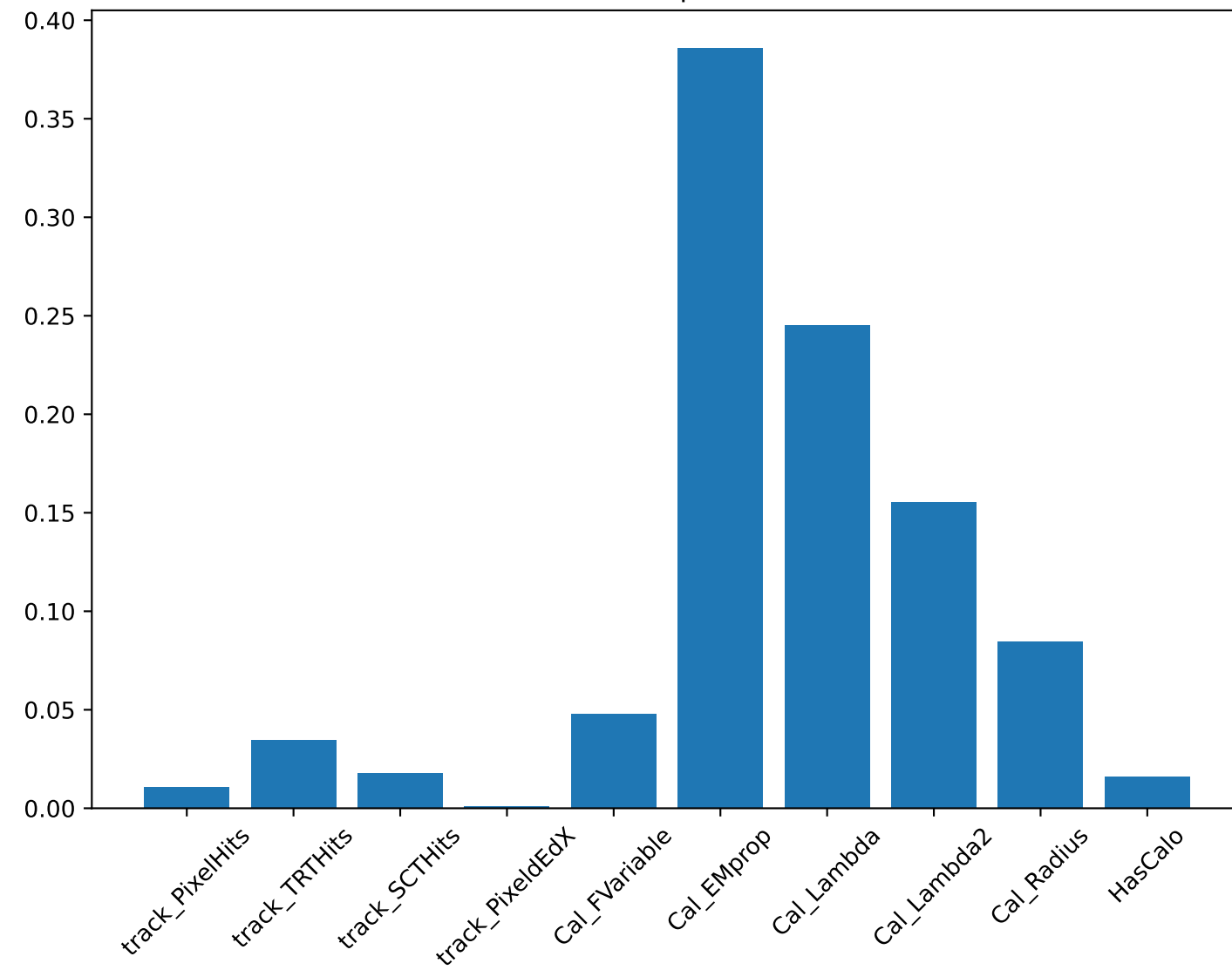
ROC Curve



Classifier Response

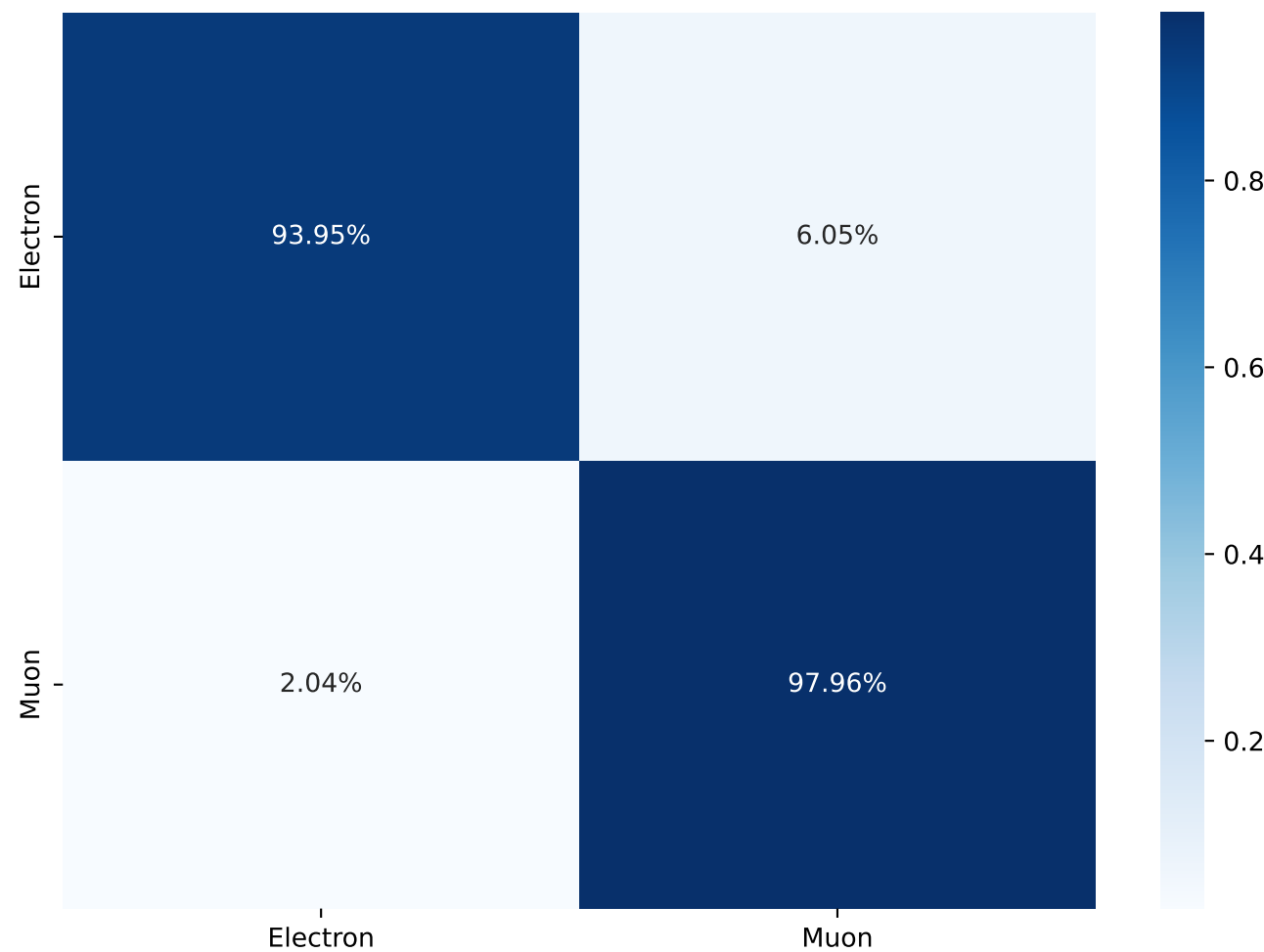


Feature Importance

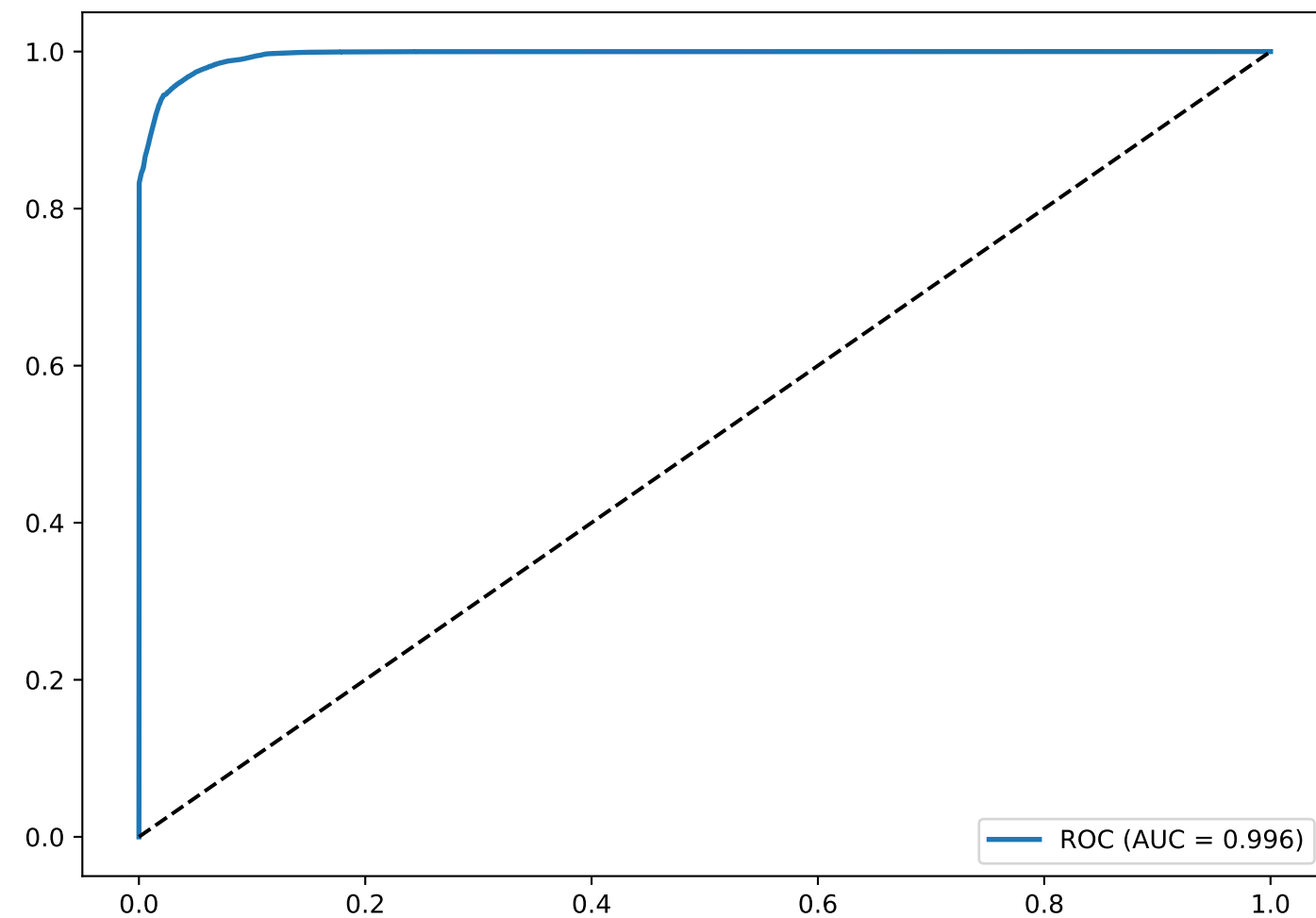


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.969

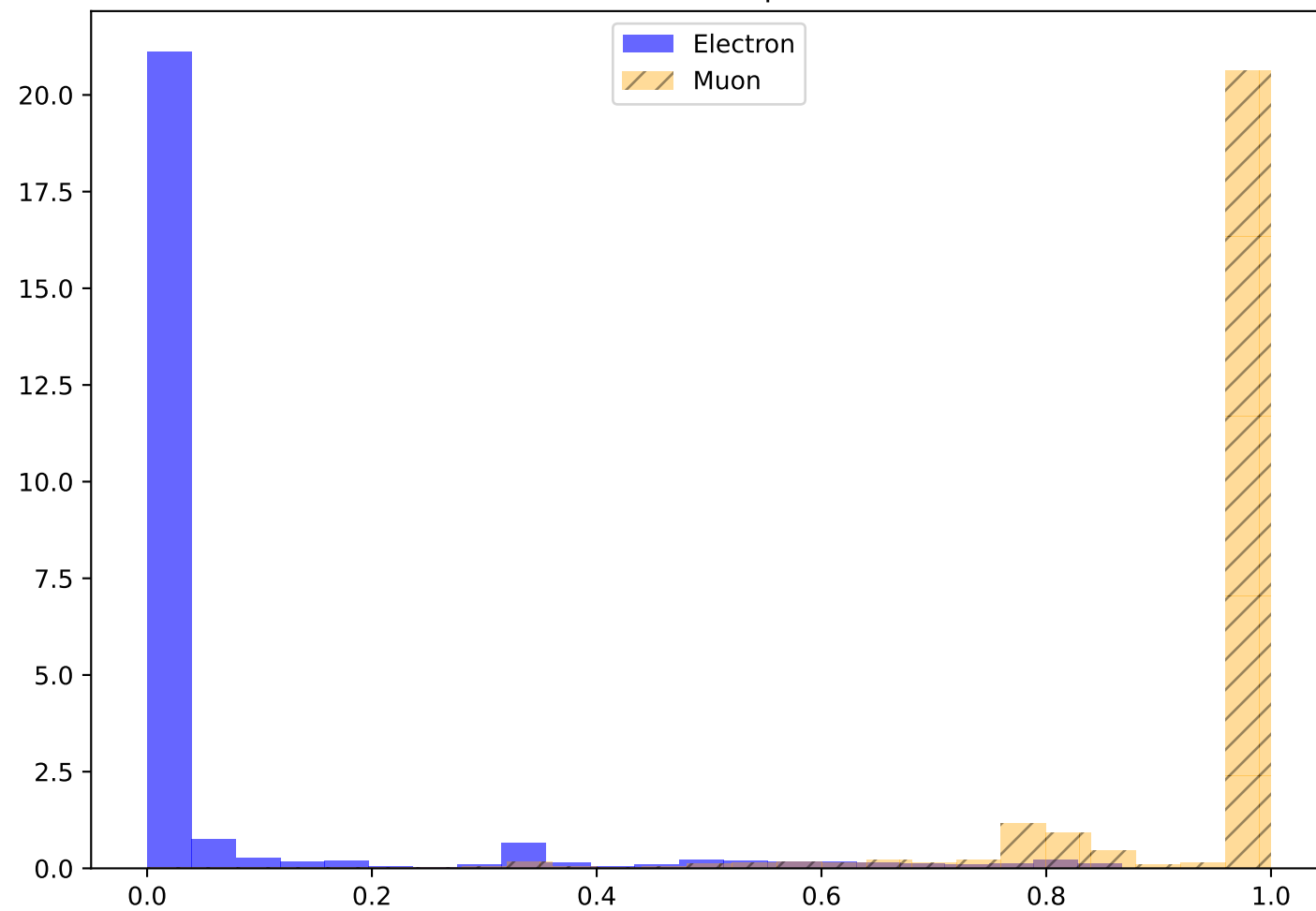
Confusion Matrix



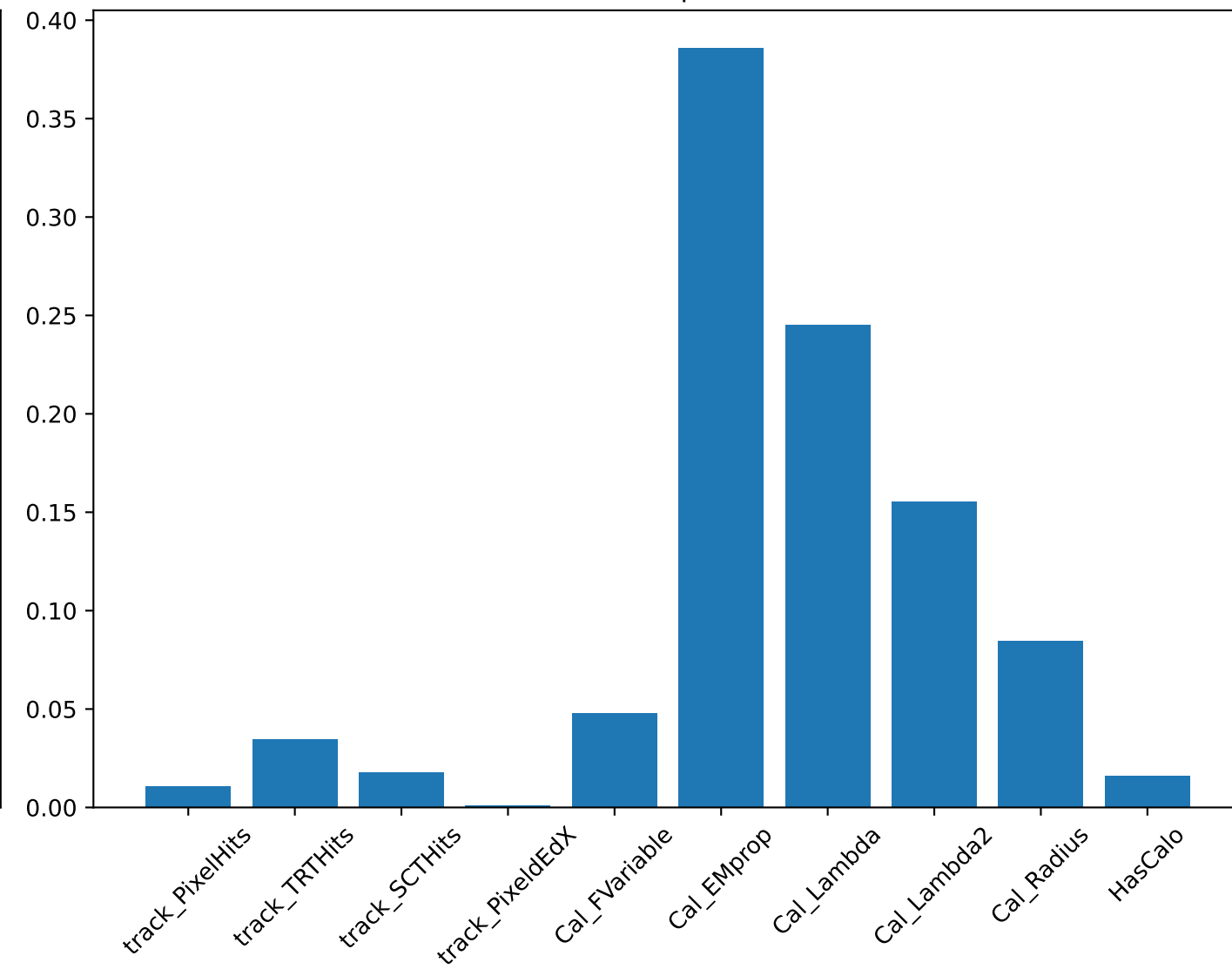
ROC Curve



Classifier Response

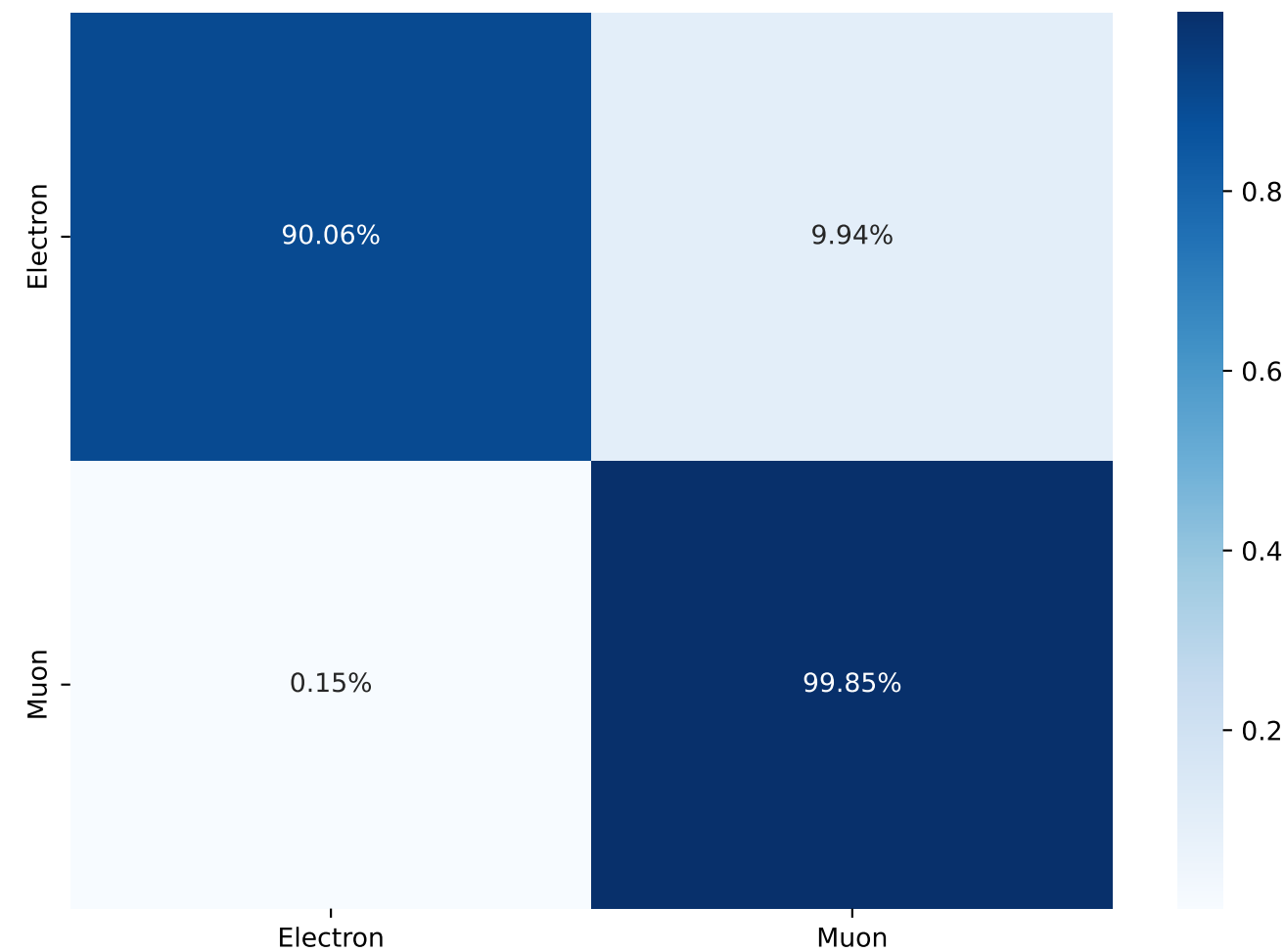


Feature Importance

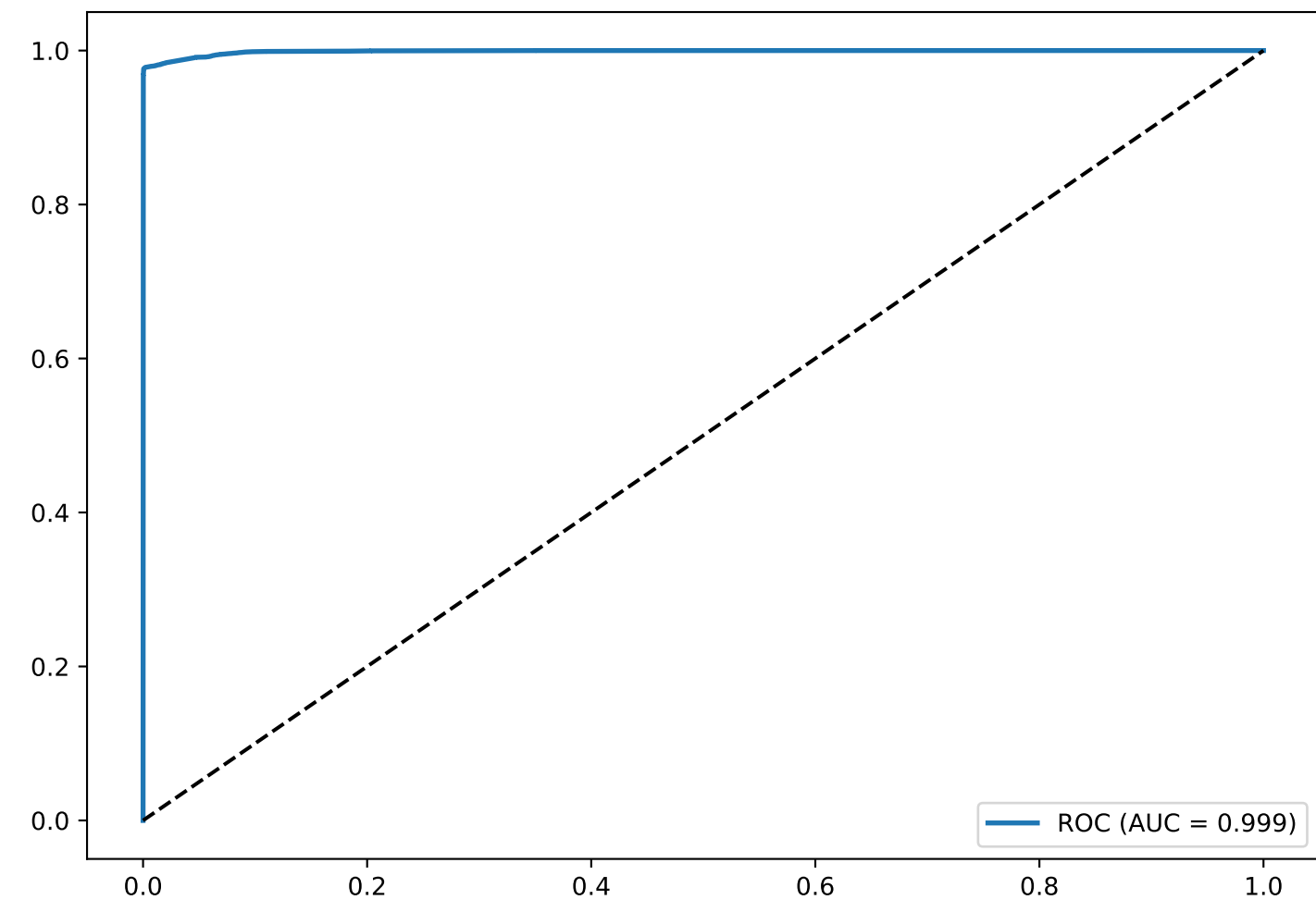


EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.977

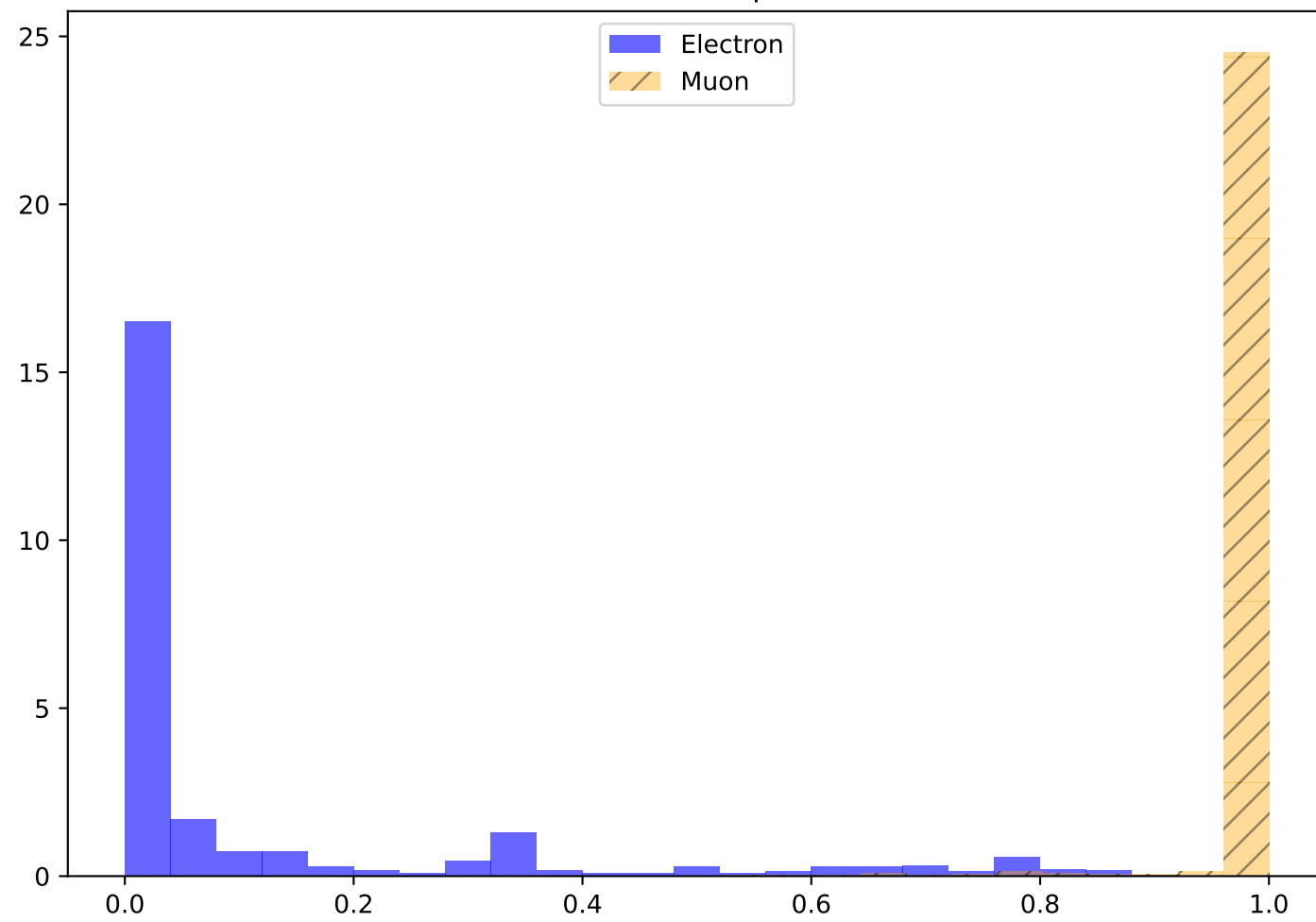
Confusion Matrix



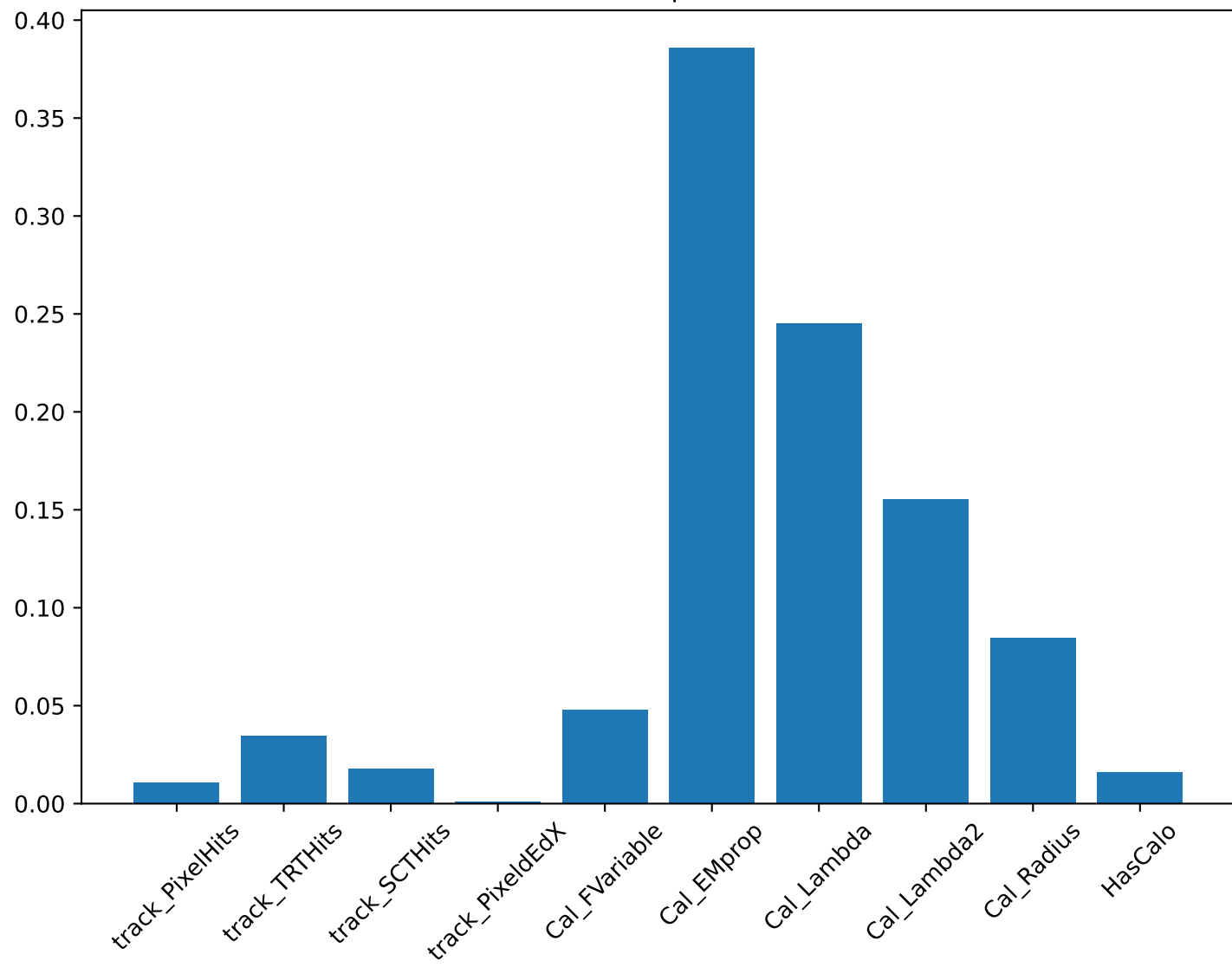
ROC Curve



Classifier Response

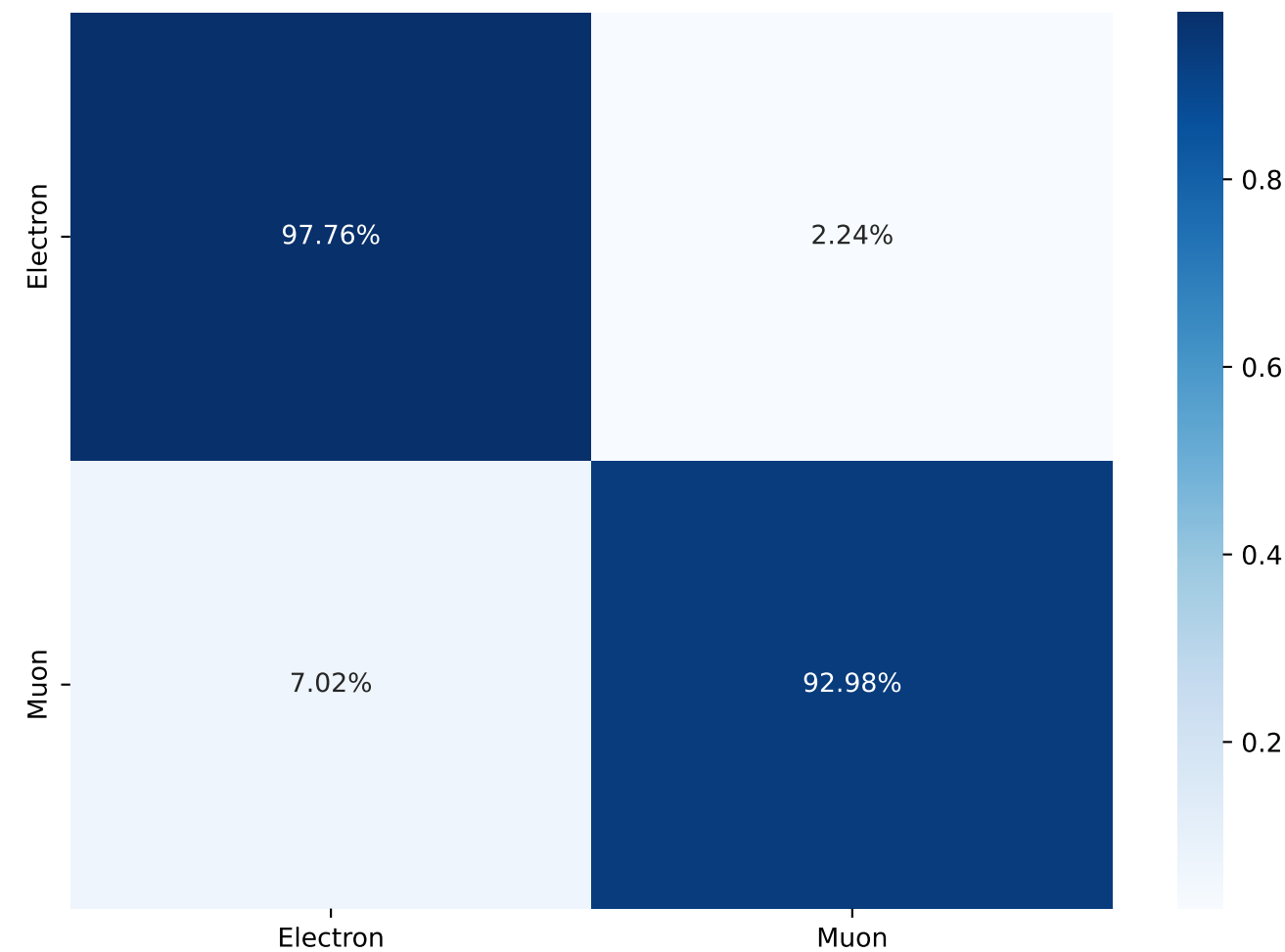


Feature Importance

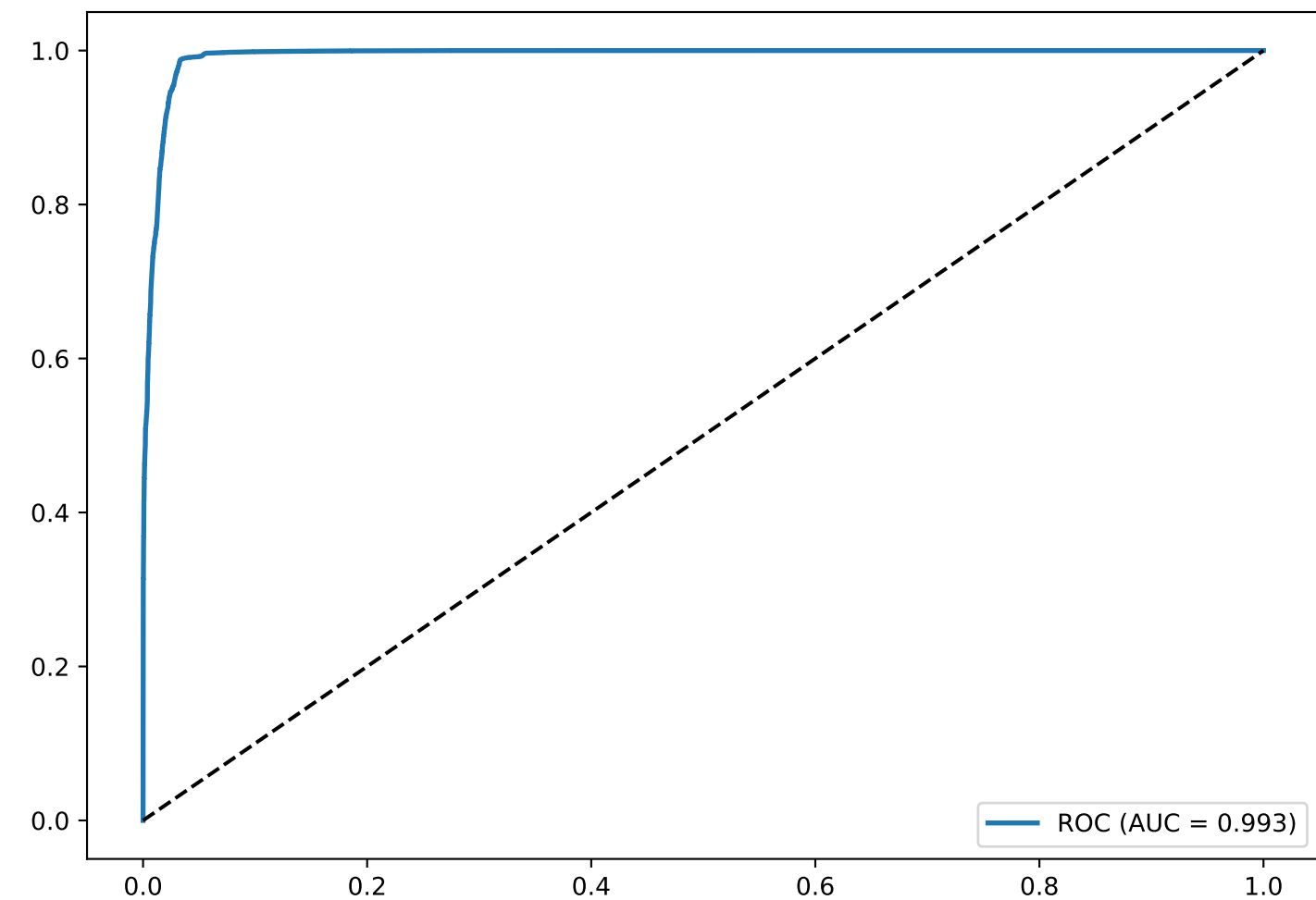


EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.936

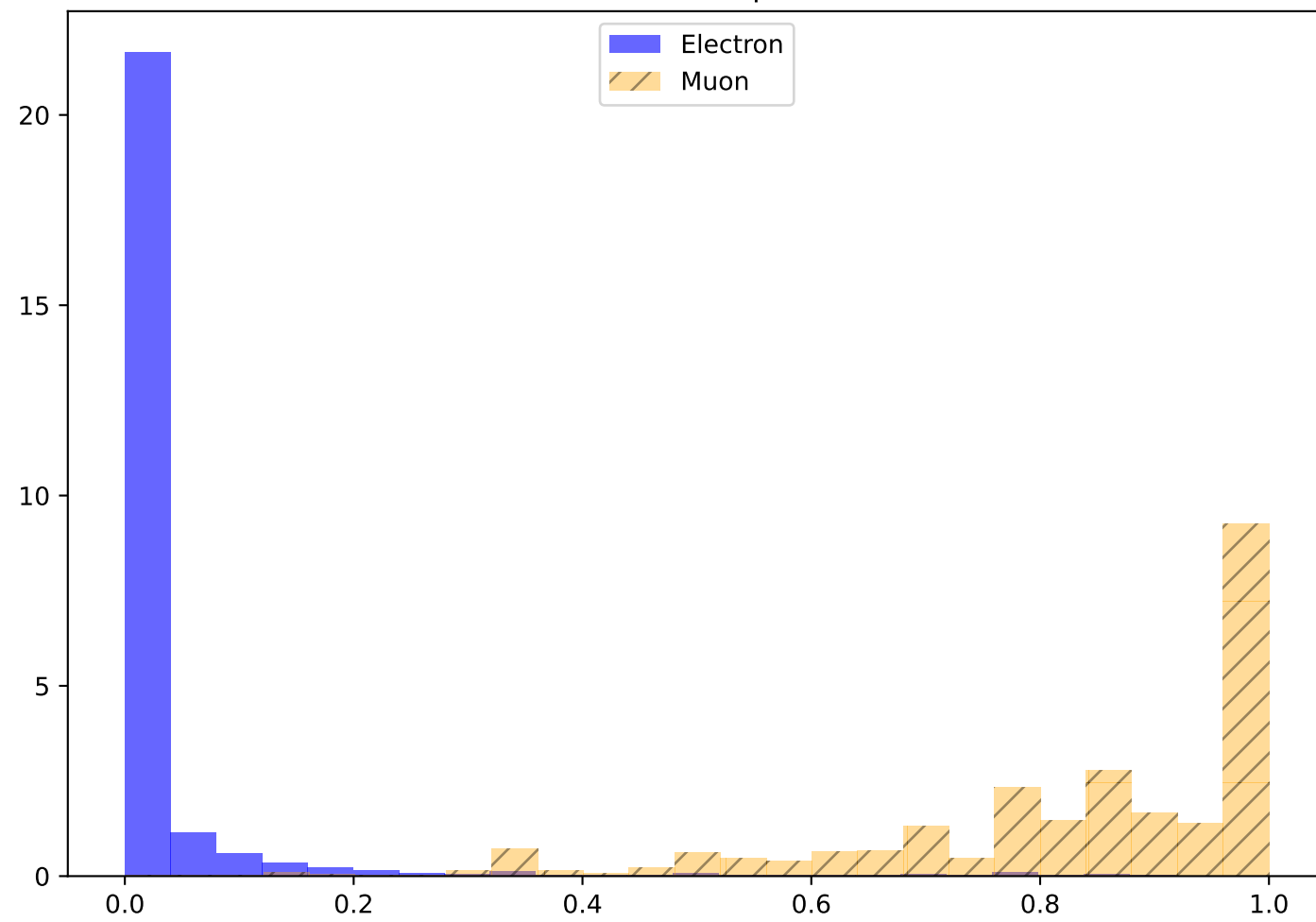
Confusion Matrix



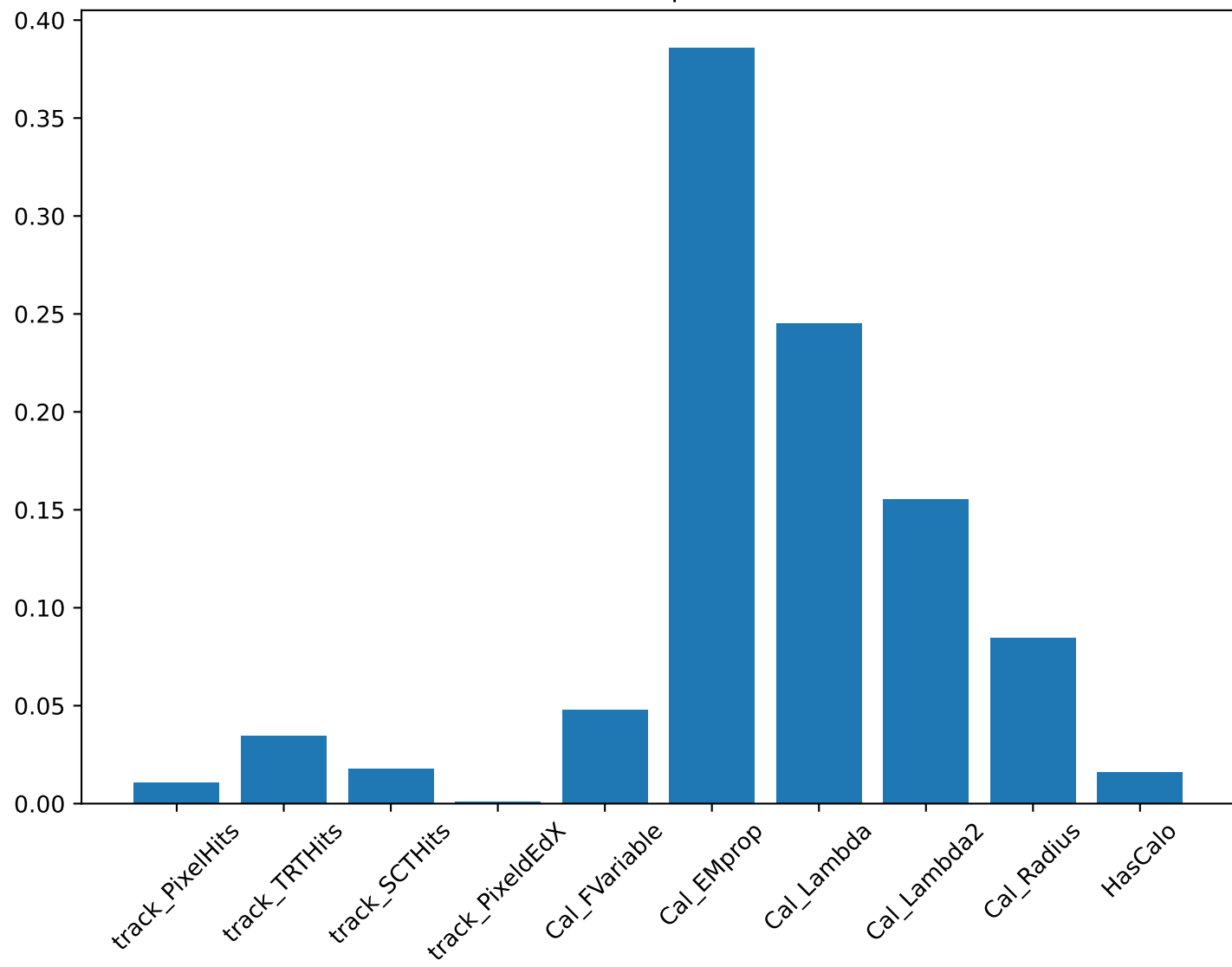
ROC Curve

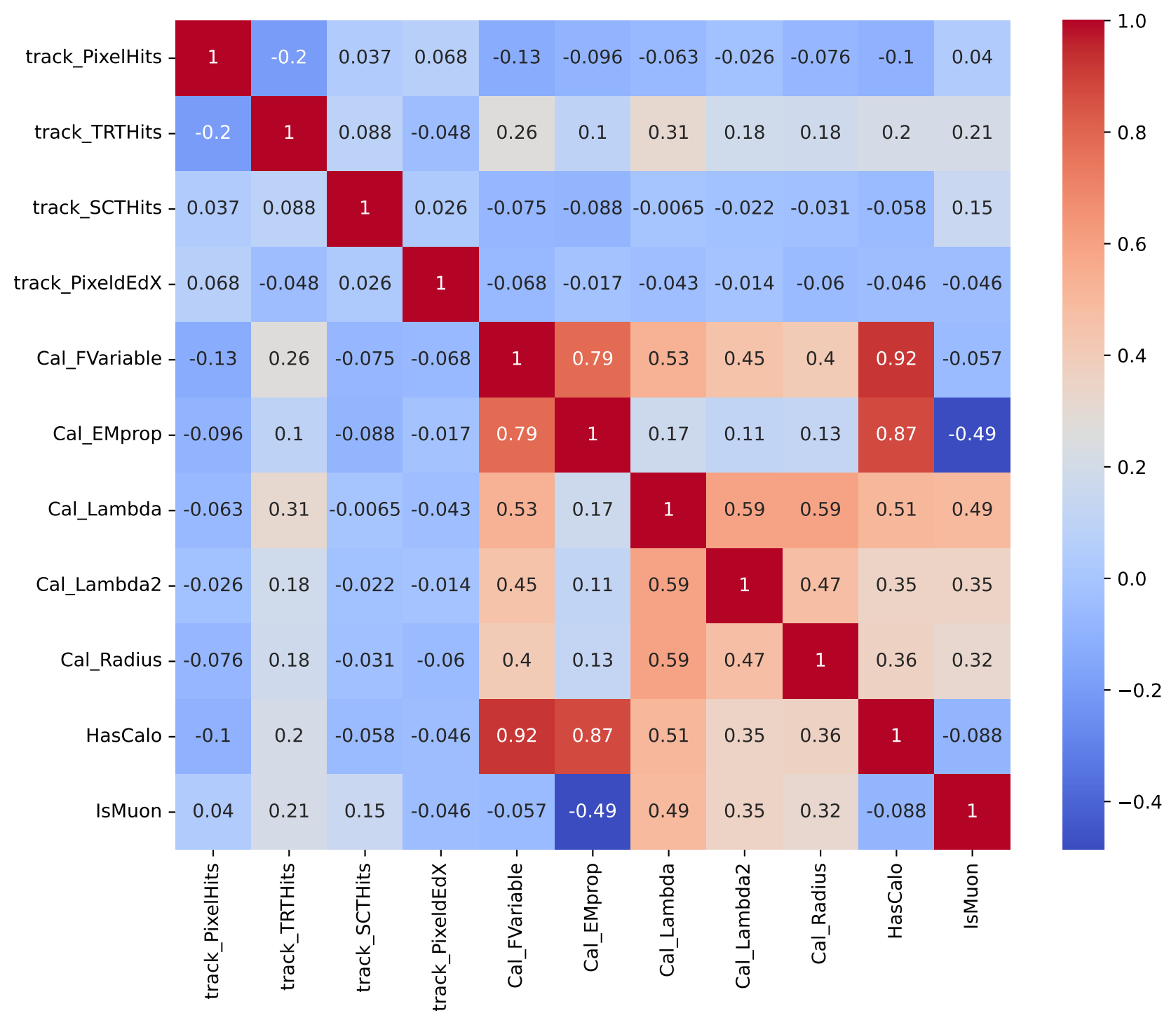


Classifier Response



Feature Importance





Classification Reports Summary

Trained: MCData.root | Test: MCData.root

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.90	0.94	0.92	102899
Muon	0.98	0.97	0.98	356159
accuracy			0.96	459058
macro avg	0.94	0.95	0.95	459058
weighted avg	0.96	0.96	0.96	459058

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.94	0.94	0.94	68367
Muon	0.98	0.98	0.98	186091
accuracy			0.97	254458
macro avg	0.96	0.96	0.96	254458
weighted avg	0.97	0.97	0.97	254458

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.90	0.95	20500
Muon	0.97	1.00	0.99	74768
accuracy			0.98	95268
macro avg	0.98	0.95	0.97	95268
weighted avg	0.98	0.98	0.98	95268

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.67	0.98	0.80	14032
Muon	1.00	0.93	0.96	95300
accuracy			0.94	109332
macro avg	0.83	0.95	0.88	109332
weighted avg	0.95	0.94	0.94	109332